

Technology Stack

Date	26 June 2026
Team ID	
Project Name	Smart Lender
Maximum Marks	2 Marks

Technology Stack:

The technology stack refers to the collection of programming languages, frameworks, libraries, and tools used to develop an application. It provides the foundation for building, testing, deploying, and maintaining the system efficiently.

A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Table below shows the technology stack used:.

Technology Stack Details:

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side (e.g., React, Angular, HTML/CSS)	<i>HTML, CSS, Javascript</i>	<i>Chosen for its fast, lightweight performance without the overhead of heavy frameworks like React. It natively integrates with Flask's templates and gave us complete control to build the responsive Premium Glassmorphism design.</i>

2	Backend / Server-Side (e.g., Node.js, Python/Django, Java)	Python / Flask	<i>Flask is an incredibly lightweight Python web framework. It was chosen because it effortlessly integrates with Python-based Machine Learning models, allowing us to load the rdf.pkl file and return real-time predictions instantly.</i>
3	Database / Data Storage (e.g., MySQL, MongoDB, PostgreSQL)	Not Applicable (Stateless Architecture)	<i>Chosen specifically to ensure 100% data privacy and security. Instead of saving sensitive financial applicant data to a database like MySQL, the system processes all inputs strictly in-memory and discards them immediately after the prediction.</i>
4	Cloud / Hosting / Deployment (e.g., AWS, Azure, Heroku, Docker)	Render	<i>The project has been deployed on Render to provide a stable and publicly accessible environment for evaluation. Unlike temporary tunneling services, Render hosts the application on a cloud server, allowing evaluators to access the project through a persistent URL without requiring the local machine to be running. This ensures better availability, reliability, and ease of access during the evaluation process.</i>
5	Version Control & CI/CD	Git & GitHub	<i>Chosen to maintain strict version control of both the Data Science pipeline (Jupyter Notebooks) and the Web Application files (app.py, style.css), ensuring secure code storage and easy rollback if errors occur.</i>

6	Third-Party APIs / Other Tools <i>(e.g., Stripe, Twilio, Google Maps)</i>	<i>XGBoost, Scikit-Learn, Pandas (Python Libraries)</i>	<i>While not a traditional web API, these are the core 3rd-party engines powering the app. Pandas handles data structuring, Scikit-Learn scales the user inputs, and XGBoost calculates the high-accuracy loan eligibility predictions.</i>
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